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High-precision temperature-strain composite sensing based on a single resonator-type surface acoustic wave sensor using machine learning algorithms

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Abstract

Surface acoustic wave (SAW) sensors are extensively employed for temperature/strain measurement. However, precise temperature/strain sensing remains challenging under complex conditions involving simultaneous mechanical strains and temperature fluctuations, due to the inherent cross-sensitivity between temperature and strain. To address this challenge, we developed a single resonator-type SAW sensor operable up to 600 °C and integrated machine learning techniques for temperature-strain multiphysics decoupling. Six machine learning models are implemented and evaluated using normalized root mean square error (*NRMSE*), root mean square error (*RMSE*), mean absolute error (*MAE*), coefficient of determination (R^2), and execution time (*ET*) metrics. Results demonstrate that the extreme gradient boosting (XGBoost) model achieves optimal predictive performance from 22 to 160 °C and applied strains of 0 $\mu\epsilon$ to 800 $\mu\epsilon$. For temperature/strain predictions respectively, R^2 values reach 99.97% and 99.23%, simultaneously enabling high-precision temperature and strain sensing. Our study presents an effective solution for simultaneous temperature and strain sensing using a single resonator-type SAW sensor operating in a single acoustic wave mode.

Introduction

Temperature/strain sensors are highly required in fields such as healthcare, aerospace, infrastructure, and energy/power systems^{1–4}. Among various measurement techniques, surface acoustic wave (SAW) sensors are widely employed for temperature/strain monitoring due to their advantages of passive wireless sensing capability, fast response speed, and high stability, etc^{5–8}. However, under real-world conditions, SAW sensors not only respond simultaneously to temperature and strain, but also exhibit

inherent temperature-strain cross-sensitivity^{9,10}. This characteristic will further amplify the interference of coupled parameter changes on measurement results, which keeps a great challenge for high-precision temperature/strain sensing⁶.

In recent years, a variety of schemes for temperature-strain multiphysics decoupling in SAW sensors are proposed. Guangyao Pei et al.¹¹ developed a SAW temperature sensor with a strain-isolation structure, which is capable of reducing cross-strain sensitivities from 176.1 to 2.5 Hz/ $\mu\epsilon$. Within the temperature range of 80–850 °C, the temperature detection error remains below 0.5‰, achieving high-precision temperature sensing performance. However, this kind of decoupling scheme completely relies on strain-isolation structure design, which leads to a complex sensor architecture and significant development challenges. For high-precision strain sensing, common decoupling schemes mainly include dual-

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resonator beat frequency and dual-mode beat frequency. The dual-resonator beat frequency scheme typically requires integrating two independent resonant units on the same substrate, and strain measurement is achieved through frequency difference calculation^{12,13}. Its structure needs to strictly control the consistency of the two resonators in the lithography process to avoid introducing additional frequency drift due to excessive deviation of the interdigital transducers. In addition, an additional isolation layer needs to be designed to reduce crosstalk between units, resulting in an increase in sensor size. More critically, this scheme often ignores the influence of temperature on strain sensitivity, thereby hindering complete temperature compensation and making it difficult to meet the requirements of high-precision scenarios^{14,15}. The dual-mode beat frequency scheme uses only a single resonator-type SAW sensor, and realizes temperature-strain decoupling through the frequency difference of exciting two different propagation modes, which significantly reduces the structural complexity¹⁶. However, not all acoustic wave modes are suitable for achieving ideal signal decoupling through beat frequency processing. Studies by Zhang et al.¹⁷ have shown that the effectiveness of dual-mode beat frequency decoupling highly depends on the consistency of the temperature response of the two modes. Only when the two modes of the same sensor have the same temperature coefficient and strain transfer ratio can the frequency difference operation offset the temperature-induced frequency drift and the strain transfer ratio variation caused by the softening of the adhesive layer. In practical applications, it is difficult to accurately match the temperature coefficients of different types of acoustic wave modes, and the excitation of multiple acoustic wave modes relies on specific structural designs, which further limits the universality of the dual-mode beat frequency scheme. In particular, this processing may usually reduce the strain sensitivity^{17,18}.

For high-precision temperature-strain composite sensing, general decoupling methods mostly involve integrating temperature-sensitive units into SAW strain sensors, or utilizing multi-resonator beat frequency. The former routine requires complete strain isolation in the temperature-sensitive unit to achieve high-precision temperature detection. Subsequently, high-precision strain detection is achieved through beat frequency processing with the SAW strain sensor or by identifying the resonant frequency-strain response curve at the specific operating temperature^{19,20}. While this decoupling method needs a complex sensor structure, a key technical challenge in practical applications remains the difficulty of achieving complete strain isolation within the temperature-sensitive unit. The latter requires three SAW resonators fabricated based on different piezoelectric materials to achieve high-precision simultaneous

measurement of temperature and strain via pairwise beat frequency processing²¹. The sensor structure is complex and ignores the influence of temperature on strain sensitivity. At present, the single acoustic wave mode-excited single resonator-type SAW sensor remains the mainstream approach, so there is an urgent need to develop a solution for its temperature-strain multiphysics decoupling.

Recently, researchers proposed to integrate machine learning techniques with sensors to achieve accurate detection of physical quantities^{22,23}. From 2022 to 2025, the research team at Hunan University conducted extensive work on the signal decoupling of SAW sensors using machine learning models. Key contributing work includes ultraviolet light-temperature decoupling¹⁰, strain direction-magnitude decoupling²⁴, precise ultraviolet detection²⁵ and temperature detection²⁶ under strain interference. However, little research was reported on temperature-strain multiphysics decoupling for SAW sensors.

In this work, we developed a high-temperature SAW sensor operable up to 600 °C, demonstrating excellent temperature sensing characteristics. Temperature/strain sensing tests were conducted under 22–160 °C and 0–800 μe loading conditions. We theoretically analyzed the temperature-strain cross-sensitivity of SAW sensors and introduced six machine learning algorithms, including ridge regression (RR), polynomial regression (PR), multiple linear regression (MLR), extreme gradient boosting (XGBoost), random forest regression (RFR), and decision tree regression (DTR), to establish regression models for temperature-strain multiphysics decoupling. Feature information was extracted from S_{11} spectra under varying temperature and strain loading conditions. Feature importance was quantified, and the impact of different input feature combinations on predictions of temperature and strain was evaluated using the normalized root mean square error (NRMSE), root mean square error (RMSE), mean absolute error (MAE), coefficient of determination (R^2) and execution time (ET) metrics. As shown in Table 1, compared with the methods reported in existing literature for decoupling SAW sensing signals based on machine learning models, the temperature-strain decoupling model constructed in this study can simultaneously achieve high-precision and accurate evaluation of both temperature and strain signals. This scheme provides a precise extraction path for temperature and strain signals for single-resonator SAW sensors excited by a single acoustic wave mode.

Fabrication and testing of high-temperature SAW sensors

Fabrication of high-temperature SAW sensors

We developed a single resonator-type high-temperature SAW sensor, primarily composed of 128° Y-cut lithium

Table 1 Comparison with the results of machine learning decoupling of surface acoustic wave sensing signals reported in the literature

Method	Target Parameter	Range	Evaluation Metrics
SVM	Temperature	30–50 °C	(Temperature) MAE: 0.25 °C, RMSE: 0.34 °C, R^2 : 99.6%
RFR ¹⁰	UV	0.033–0.403 mW/cm ²	(UV) MAE: 0.011 mW/cm ² , RMSE: 0.0136 mW/cm ² , R^2 : 95.9%
XGBoost ²⁴	Strain intensity	0–1888 $\mu\epsilon$	(Strain intensity) RMSE: 0.035 $\mu\epsilon$, R^2 : 98.6%
	Strain direction	0–90 deg	(Strain direction) RMSE: 0.054 deg, R^2 : 98%
XGBoost ²⁵	UV	0–13.05 mW/cm ²	R^2 : 99.6%
	Strain	0–1160 $\mu\epsilon$	NRMSE: 0.019
XGBoost ²⁶	Temperature	0–80 °C	NRMSE < 1%
	Off-axis bending strain	0–1000 $\mu\epsilon$	R^2 > 99.7%
XGBoost (This work)	Temperature	22–160 °C	(Temperature) R^2 : 99.97%, NRMSE: 0.0059,
	Strain	0–800 $\mu\epsilon$	RMSE: 0.82 °C, MAE: 0.35 °C (Strain) R^2 : 99.23%, NRMSE: 0.0283, RMSE: 22.65 $\mu\epsilon$, MAE: 8.42 $\mu\epsilon$

Table 2 Structural parameters of the high-temperature SAW sensor working in the ISM frequency band

Structural parameters	Value
Wavelength, λ (μm)	8.48
Metallization ratio	0.5
Acoustic aperture (mm)	0.848
Thickness of LN (mm)	0.5
The number of IDT electrode pairs	60
The number of RG pairs	60
Distance between IDT and RG (μm)	2.12

niobate (LN), interdigital transducer (IDT), reflective gratings (RG) at both ends, and an aluminum nitride (AlN) protective layer. To enable the first-order Rayleigh SAW (RSAW) of the sensor to operate within the commercial ISM (industrial, scientific, and medical) bands, the period of IDT was set to 8.48 μm based on previous research results²⁷, with detailed structural parameters provided in Table 2.

The fabrication process flow of the high-temperature SAW sensor is shown in Fig. 1a, mainly including dicing, magnetron sputtering deposition of metal electrode films, photolithographic patterning, etching, deposition of an AlN protective layer, and post-annealing. To meet high-temperature testing requirements, we selected chromium (Cr) and platinum (Pt) as electrode materials for the high-temperature SAW sensor due to their stability and high electrical conductivity in the high-temperature conditions. Detailed process parameters for the deposition of Cr and Pt metal films are listed in Table S1 of the

Supporting Information. Subsequently, we spin-coated a layer of photoresist onto the metal thin film and transferred the pre-designed electrode pattern onto the photoresist layer via photolithography. Next, ion beam etching was performed using the patterned photoresist as a mask. After stripping off the residual photoresist, the electrode morphology of the SAW sensor was characterized, with the results presented in Fig. 1b. This image clearly demonstrates that the electrodes exhibit well-defined profiles, uniform width and no abnormal interconnections.

However, the SAW sensor electrodes are likely to be oxidized in the high-temperature environment, and may even exhibit recrystallization and agglomeration phenomena, leading to IDT fracture and subsequent sensor failure. Additionally, Li^+ volatilization in LN further degrades performance of the sensor. In this work, we masked the Pad of the high-temperature SAW sensor by applying ink and deposited the AlN protective layer using the DC magnetron sputtering process. The specific process parameters are listed in Table S2 of the supplementary material. With a melting point of approximately 2200 °C, AlN exhibits high thermal stability, excellent compactness, electrical insulation, and chemical stability, enabling effective protection of metal electrodes and the LN piezoelectric substrate in the high-temperature environment. Furthermore, as an excellent piezoelectric material, AlN maintains stable piezoelectric performance even at 1150 °C²⁸. Therefore, it serves both as a protective layer and enhances electromechanical coupling to improve SAW sensor performance. The cross-sectional scanning electron microscopy (SEM) image of the high-temperature SAW sensor is shown in Fig. 1c, clearly revealing that the deposited Cr and Pt metal electrode

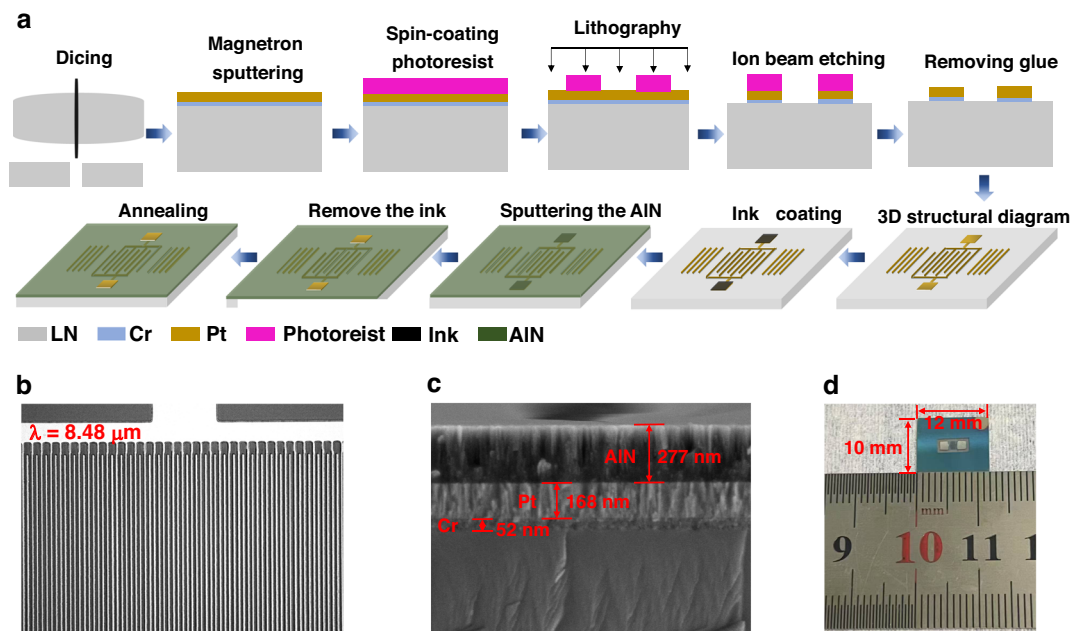


Fig. 1 Fabrication Process and Morphological Structure of High-Temperature SAW Sensor. **a** The fabrication process flow; **b** The metallographic microscope image of the electrode surface; **c** The cross-sectional SEM image and **d** the physical drawing of the high-temperature SAW sensor

films have thicknesses of 52 nm and 168 nm, respectively, and the AlN protective layer has a thickness of 277 nm.

After all the preparation processes are completed, the sample is annealed in a nitrogen atmosphere. The annealing process involves heating the sample to 600 °C at a rate of 5 °C/min, holding it at this temperature for 30 min, and then cooling it naturally to room temperature. This treatment improves the interface quality between the electrode material and the substrate, promotes the recrystallization of the metal thin-film electrode to enhance conductivity, and eliminates the internal stress accumulated in the thin film during fabrication, thereby improving the performance, service life, and operational stability of the sensor. The photograph of the prepared high-temperature SAW sensor is shown in Fig. 1d.

The temperature-strain composite sensing test system

The temperature-strain composite sensing test system is shown in Fig. 2, comprising of the following components, including the high-temperature SAW sensor, an extensometer, a test piece, a tensile testing machine, a high-temperature furnace, a network analyzer (E5071B, Keysight, USA), and a computer. The tensile testing machine applies strain to the test piece, while the high-temperature furnace controls the test temperature. During testing, the high-temperature SAW sensor is bonded to the test piece using adhesive, then electrically connected to a printed circuit board (PCB), ultimately interfacing with the

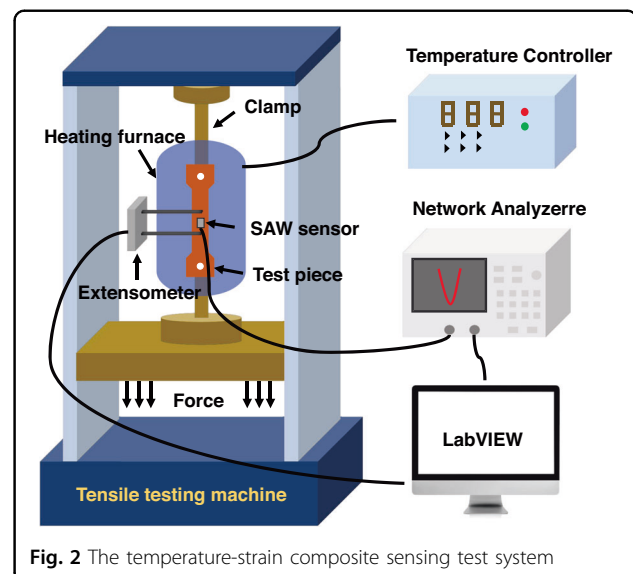
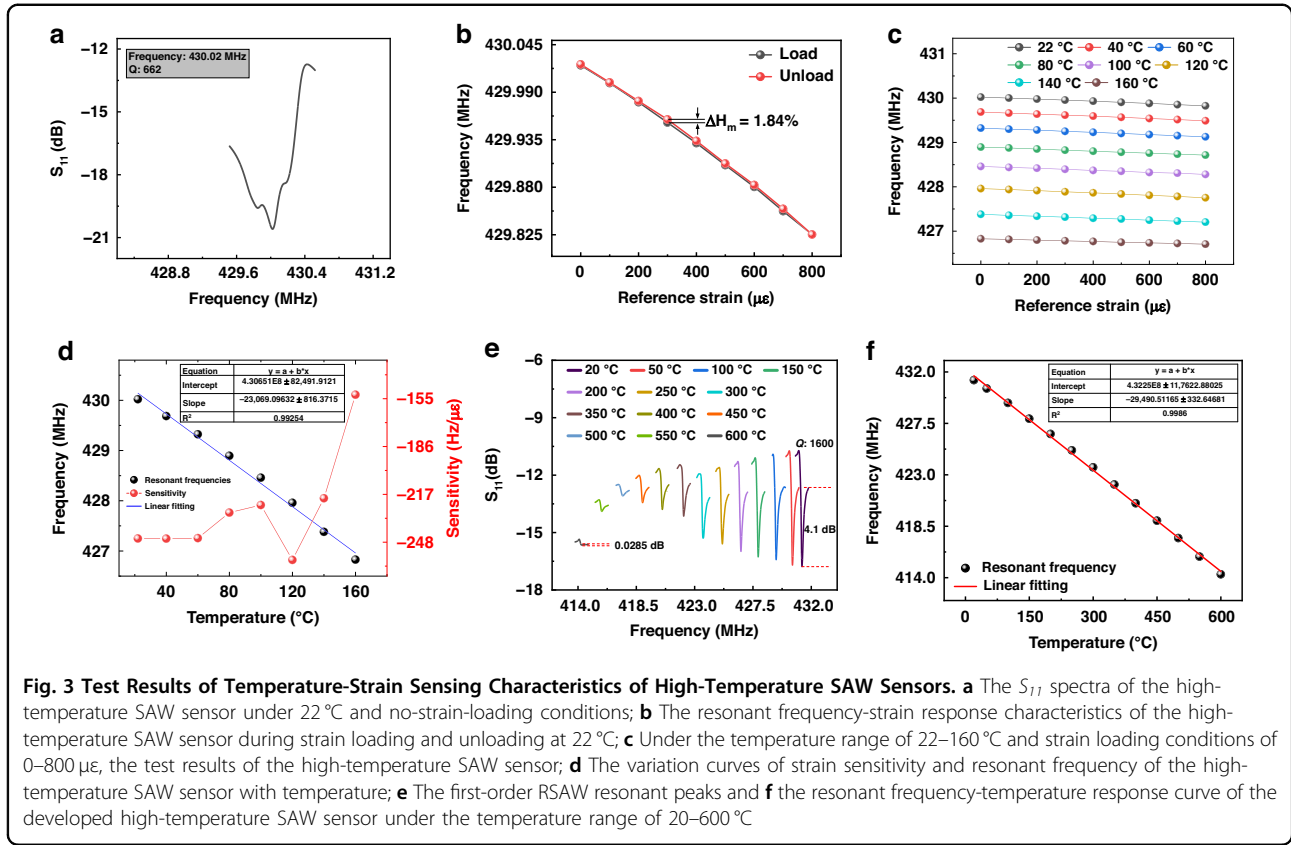


Fig. 2 The temperature-strain composite sensing test system

network analyzer through an SMA connector on the PCB. An EFSS self-clamping axial high-temperature extensometer with a 50 mm gauge length, fixed at both ends of the SAW sensor, calibrates strain of the test piece, and its measurement error is $\pm 0.15\%$. Test temperature is calibrated using a thermocouple with a measurement error of $\leq \pm 1$ °C. The network analyzer characterizes the S_{11} spectra of the SAW sensor, which connects to the



computer via data cables. Automated measurement of S_{11} parameters and resonant frequency extraction are achieved through LabVIEW-based programming.

Analyses of temperature/strain sensing characteristics

The fabricated high-temperature SAW sensor was bonded to the test piece using adhesive glue (3M-CA40H) and electrically connected to the PCB via twisted-pair wires using conductive silver paste and solder paste for testing of temperature/strain sensing characteristic. The S_{11} spectrum of the sensor under room temperature (22 °C) and no-strain-loading conditions are shown in Fig. 3a. A distinct resonant peak is observed at 430.02 MHz, exhibiting good quality with a quality factor (Q) of 662, which indicates that the developed high-temperature SAW sensor has excellent resolution and a longer wireless transmission distance. The Q is calculated using the following formula²⁹:

$$Q = \frac{f_0}{B_w} \quad (1)$$

where f_0 is the resonant frequency and B_w is the bandwidth at 3 dB. A minor deviation exists between the resonant frequency and the theoretically calculated 433 MHz, which may be attributed to the influence of the

AlN protective layer on the propagation velocity of the first-order RSAW. The resonant frequency-strain response characteristics of the high-temperature SAW sensor during strain loading and unloading at 22 °C are shown in Fig. 3b. It demonstrates ideal strain sensing characteristics with high strain sensitivity (−245.63 Hz/ $\mu\epsilon$) and linearity (99.8%). Additionally, it exhibits a low maximum hysteresis error (ΔH_m) of 1.84%. ΔH_m denotes the maximum deviation of the output resonant frequency during strain loading and unloading within the testing range, and is defined as³⁰:

$$\Delta H_m = \frac{\Delta f_m}{Y_{FS}} \times 100\% \quad (2)$$

where Δf_m is the maximum deviation of the resonant frequency at the same strain, and the Y_{FS} is the resonant frequency range of the sensor output. Within the temperature range of 22–160 °C and strain loading conditions of 0–800 $\mu\epsilon$, the test results of the high-temperature SAW sensor are shown in Fig. 3c. It shows that the resonant frequency versus strain exhibits excellent linear response characteristics across temperatures, with a linearity range of 99.8–99.91%. This indicates its high strain detection accuracy and minor errors in repeated measurements. Based on the test data shown in

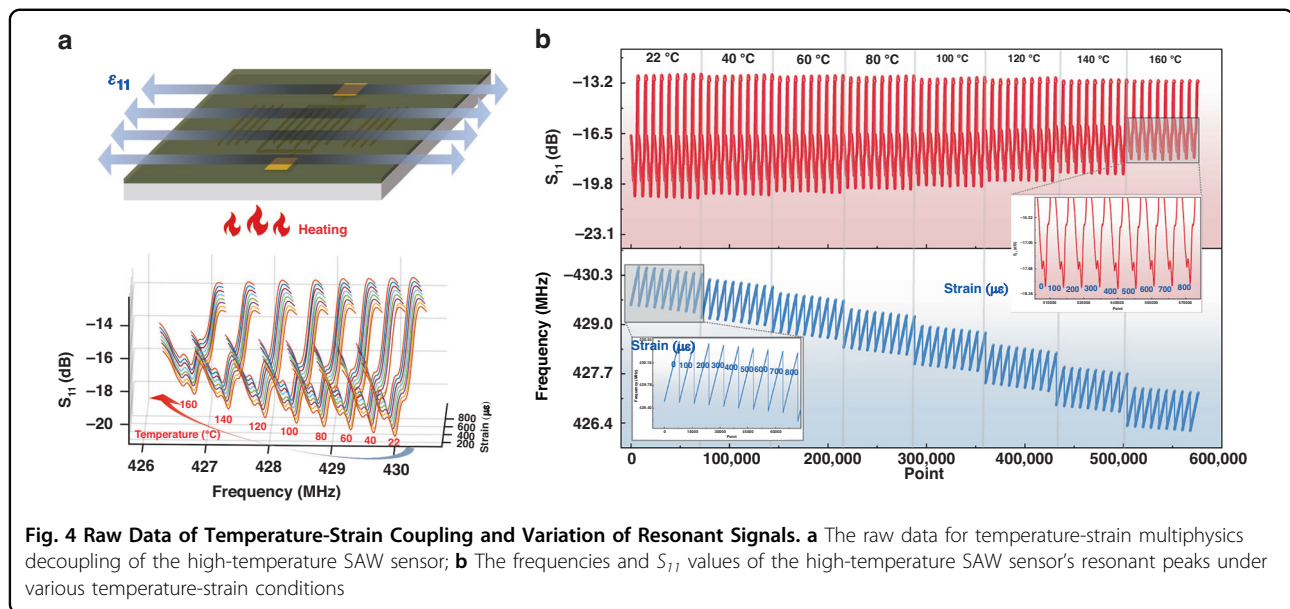


Fig. 3c, we extracted the variation curves of strain sensitivity and resonant frequency of the sensor with temperature, and the results are shown in Fig. 3d. It shows that the strain sensitivity exhibits strong temperature-dependent fluctuations, while the resonant frequency-temperature response curve demonstrates linear variation characteristics with approximately 99.3% linearity and a sensitivity of $-23069 \text{ Hz}/^\circ\text{C}$. This also demonstrates the cross-sensitivity characteristics of the SAW sensor to temperature and strain. Therefore, it needs signal decoupling to obtain accurate temperature and strain information.

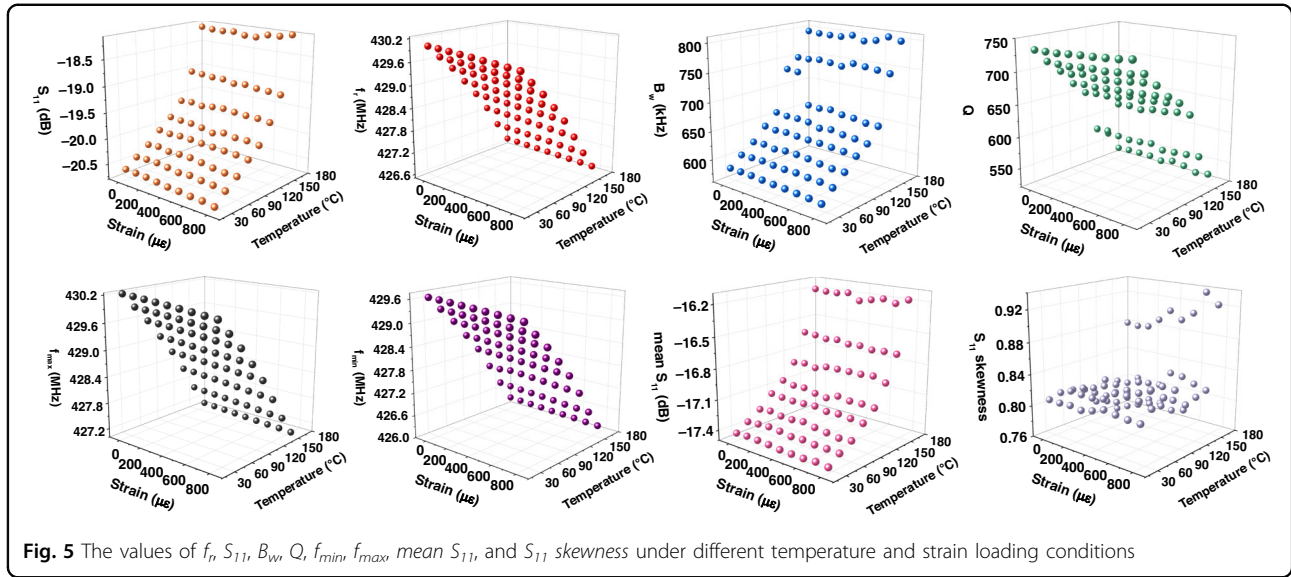
To validate the temperature/strain sensing characteristics of the high-temperature SAW sensor under higher environmental temperatures, we adopt new electrical connection and installation processes. First, the high-temperature SAW sensor was bonded to a TC4 test piece using ceramic adhesive (GT-900-H). Then, it was electrically connected to a PCB via a dual-core thermocouple wire using high-temperature conductive silver paste. The dual-core thermocouple wire is long enough to ensure that the electrical signals from the sensor can be routed out of the high-temperature environment, allowing the PCB to remain in the room-temperature zone. Figure 3e shows the first-order RSAW resonant peaks of the developed high-temperature SAW sensor under controlled temperatures from 20 to 600°C using a heating stage. As temperature increases, the amplitude of resonant peak gradually decreases and the resonant peak gradually broadens, reaching only 0.0285 dB at 600°C . Additionally, the temperature-dependent resonant frequencies extracted from these resonant peaks are linearly fitted, as shown in Fig. 3f, achieving a temperature

sensitivity of $-29,490.5 \text{ Hz}/^\circ\text{C}$ with 99.86% linearity. This result confirms the sensor's operational capability and excellent temperature sensing performance within $20\text{--}600^\circ\text{C}$. However, validation of its strain sensing capability remains incomplete due to insufficient strain transfer through the ceramic adhesive and a 0.5 mm-thick LN substrate. Therefore, subsequent temperature-strain multiphysics decoupling is conducted based on data obtained under $22\text{--}160^\circ\text{C}$ and $0\text{--}800 \mu\epsilon$ strain loading conditions.

Decoupling methods using machine learning models

Temperature-strain cross-sensitivity characteristics in SAW sensors

The raw data for temperature-strain multiphysics decoupling of the high-temperature SAW sensor is shown in Fig. 4a. The temperature test points include 8 values: 22°C , 40°C , 60°C , 80°C , 100°C , 120°C , 140°C , and 160°C . The strain test points consist of 9 values: $0 \mu\epsilon$, $100 \mu\epsilon$, $200 \mu\epsilon$, $300 \mu\epsilon$, $400 \mu\epsilon$, $500 \mu\epsilon$, $600 \mu\epsilon$, $700 \mu\epsilon$, and $800 \mu\epsilon$. This configuration results in 72 distinct target variables. Each target variable corresponds to a frequency response dataset containing 8000 data points centered at the resonant frequency. The data points are sampled at intervals of 125 Hz, with 4000 points symmetrically extended on each side of the resonant frequency, resulting in a total bandwidth of 1 MHz. Collectively, the 72 target variables comprise 576,000 data points, forming the foundational dataset for machine learning-based temperature-strain multiphysics decoupling. Figure 4b illustrates the resonant peak frequency and S_{11} value of the high-temperature SAW sensor under different



temperature-strain loading conditions. These parameters vary regularly with temperature and strain, exhibiting a distinct coupling effect. When temperature rises, the resonant frequency decreases, and simultaneously, the frequency-reducing effect induced by increasing strain becomes more pronounced with elevated temperatures. The amplitude variation of S_{11} is the combined result of temperature and strain. Higher temperatures attenuate the S_{11} amplitude, whereas larger strain levels enhance it, ultimately yielding fluctuation characteristics that stem from the superposition of these two factors. Specifically, the resonant frequency decreases with increasing temperature and strain, while the amplitude of $|S_{11}|$ declines as temperature rises but generally increases with growing strain. This phenomenon primarily arises from the fact that both temperature and strain induce alterations in the IDT period of SAW sensors, thereby modifying the wavelength of SAW. Additionally, these factors also affect parameters such as the density and elastic modulus of piezoelectric materials³¹, leading to changes in SAW propagation velocity. These combined effects ultimately lead to the changes in the resonant frequency, which cause shifts in the impedance matching point, ultimately affecting the S_{11} values. In summary, both temperature and strain affect the output frequency signals of SAW sensors, demonstrating significant coupling effects between the two parameters.

Feature extraction

Machine learning models analyze training data through specified algorithms to uncover latent patterns and relationships, which are then used to predict new data³². In this work, machine learning models are employed to predict the temperature and strain

information detected by the high-temperature SAW sensor, where each input dataset corresponds to a known output target variable (i.e., the calibration temperature and strain). The objective of the models is to identify the underlying relationships between input data and output target variables, which typically requires first extracting features from the raw input data to establish input-output mapping relationships³³. However, only two features can be directly obtained from the SAW sensor's resonant peaks: resonant frequency (f_r) and its corresponding S_{11} value, which are insufficient for constructing effective input-output mappings relationships to achieve temperature-strain multiphysics decoupling. To further enhance the prediction accuracy of machine learning models, we extract six new features based on the frequency characteristics of the high-temperature SAW sensor's resonant peaks, including bandwidth (B_w), Q , minimum cutoff frequency (f_{min}), maximum cutoff frequency (f_{max}), $mean\ S_{11}$, and $S_{11}\ skewness$. Their values under different temperature-strain loading conditions are shown in Fig. 5. As shown in Fig. 5, the variation trends of the cutoff frequencies (f_{min} , f_{max}) are fully synchronized with that of the f_r . Specifically, an increase in temperature or strain induces a low-frequency shift of f_r , which consequently drives a synchronous decrease in both f_{min} and f_{max} . The difference between the two is the bandwidth B_w . Among them, high temperature will increase the internal friction of the piezoelectric material, resulting in the broadening of B_w and the decrease of the Q ; while the influence of strain on B_w is very weak, which only causes small fluctuations in the Q value by changing the propagation uniformity of SAW. The $mean\ S_{11}$ provides a global measure of S_{11} values, reflecting the energy attenuation characteristics of the SAW sensor

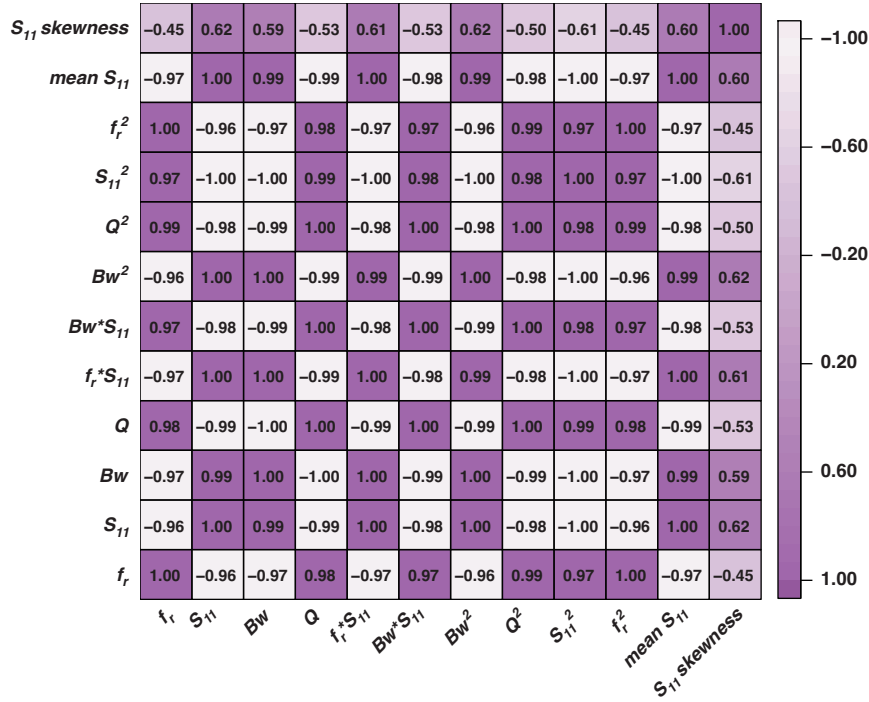


Fig. 6 Pearson correlation coefficient matrix of extracted features

under specific conditions. It is defined as:

$$mean S_{11} = \frac{1}{n} \sum_{i=1}^n S_{11i} \quad (3)$$

where n represents the total number of datasets, and S_{11i} denotes the S_{11} value of the i -th dataset. Compared with the S_{11} value of a single resonance peak, the mean S_{11} can suppress noise interference in the original signal and more accurately reflect the global energy loss caused by temperature. In terms of numerical changes, the mean S_{11} decreases significantly as the temperature rises. The S_{11} skewness is a statistical metric describing the symmetry of data distribution, defined as³⁴:

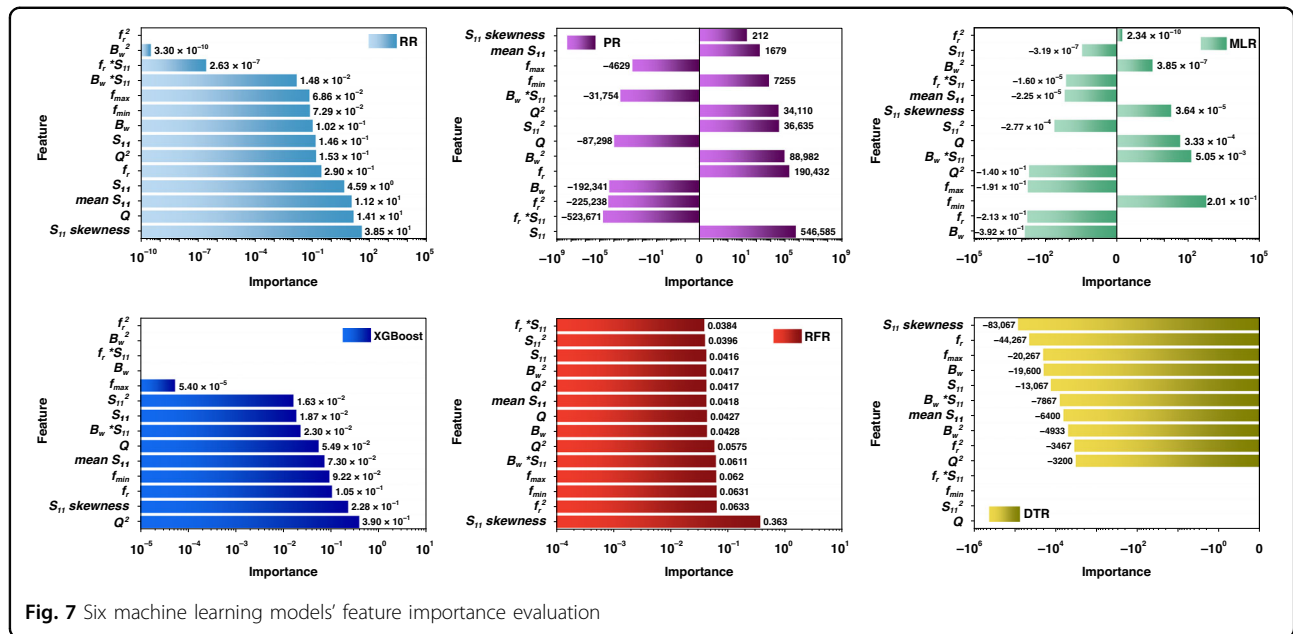
$$S_{11} \text{ skewness} = \frac{\frac{1}{n} \sum_{i=1}^n (S_{11i} - \overline{S_{11}})^3}{\left(\frac{1}{n} \sum_{i=1}^n (S_{11i} - \overline{S_{11}})^2 \right)^{3/2}} \quad (4)$$

When under lower temperatures and smaller strain loads, the S_{11} spectrum shows a symmetric normal distribution, with the skewness close to 0. However, when the temperature rises or the strain increases, the impedance matching state of the sensor becomes imbalanced, causing the spectrum to shift in a certain direction, which in turn leads to positive and negative fluctuations in skewness. This

feature can capture subtle asymmetric changes in the original signal that are difficult to observe directly, providing additional basis for judging the temperature-strain coupling state. Additionally, six polynomial product-form features are introduced in this work, including Bw^2 , Q^2 , f_r^2 , S_{11}^2 , $f_r S_{11}$ and $Bw S_{11}$. These additions significantly improve the model's predictive performance.

Feature importance evaluation

After performing feature extraction on the original input data, it is necessary to further verify whether there is redundancy or multicollinearity among the features. The Pearson correlation coefficient matrix plot, which reflects the linear correlation between features, is shown in Fig. 6. The value range of the correlation coefficient is $[-1, 1]$; the closer the absolute value is to 1, the stronger the linear correlation between features. A positive value indicates a positive correlation, and a negative value indicates a negative correlation. It can be seen from Fig. 6 that the absolute values of the correlation coefficients for most feature pairs are close to 1, showing a significant linear correlation. Among them, the strong correlation between the original features and their own polynomial terms is an inherent mathematical relationship in polynomial expansion, while the strong correlation between different original features reflects the linear dependence of the features themselves. However, the impact of such strong correlations on subsequent analysis varies depending on



the type of model. For linear models (RR, PR, MLR), multicollinearity may interfere with the stability of coefficient estimation, and the subsequent ranking of feature importance is more dependent on the strength of linear correlation. For tree-based models (XGBoost, RFR, DTR), by contrast, they are less sensitive to multicollinearity, and polynomial features can just help capture the nonlinear relationships between temperature/strain and SAW signals. This in turn can provide richer information for the model and is crucial for improving prediction accuracy.

To quantify the contribution of individual input feature labels to the predictive performance of the models, this study calculated and ranked the feature importance of six machine learning models (RR, PR, MLR, XGBoost, RFR, DTR). The detailed calculation process is provided in the supplementary materials, and the results are shown in Fig. 7. When constructing machine learning models, prioritizing high-importance features while removing low-importance ones retains critical variables, reduces noise interference, and enhances computational efficiency and model generalization capability¹⁰. Due to the SAW sensor's higher sensitivity to temperature, temperature exerts significant interference on strain detection. For the resonant frequency, its temperature sensitivity is two orders of magnitude higher than its strain sensitivity. Therefore, for temperature-strain multiphysics decoupling, precise temperature information becomes more readily extractable. To enable machine learning models to achieve accurate predictions for both temperature and strain, we designated strain as the output target variable.

As can be seen from Fig. 7 there are significant differences in the feature importance distribution among the 6 machine learning models. Linear models focus

more on features that have a linear correlation with the target variable, and the feature importance is measured by the absolute value of the coefficient. The value distribution is relatively concentrated without extreme differences. This is because linear models can only capture the linear relationship between features and the target and cannot handle complex nonlinear interactions. Therefore, they pay less attention to polynomial product-type features, and only PR assigns a certain weight to some product features due to the introduction of polynomial terms. Tree-based models exhibit a more scattered distribution of feature importance and attach greater importance to statistical or nonlinear features such as S_{11} skewness and Q^2 . This is because tree-based models achieve iterative optimization through node splitting, which can independently explore nonlinear correlations and complex interaction effects between features. They are more sensitive to features that reflect data distribution characteristics (S_{11} skewness) or high-order correlations (such as square terms and product terms). In addition, because the DTR model has a single-tree structure, the model's selection of features is more direct, which leads to significant fluctuations in the importance of features. In contrast, XGBoost and RFR reduce the dependence on a single feature by integrating multiple trees, resulting in a relatively more balanced distribution of feature importance, but they still focus primarily on nonlinear features. This difference essentially stems from the algorithmic logic of different models: linear models rely on the linear correlation between features and the target, while tree-based models rely on the ability of features to reduce the impurity of data partitioning. Consequently, the criteria for

determining the importance of input features are completely different.

Evaluation metrics of model performance

We employed three evaluation metrics for comprehensive performance assessment, namely *ET*, *NRMSE*, *MAE*, *RMSE* and R^2 . The *NRMSE* is a standardized measure quantifying the error magnitude between predicted and calibration values in regression models. A smaller *NRMSE* value denotes higher predictive accuracy of the model, with predicted values closer to the calibration values. The definition of *NRMSE* is as follows²⁶:

$$NRMSE = \frac{\sqrt{\frac{1}{l} \sum_{i=1}^l (y_i - \hat{y}_i)^2}}{\max(y) - \min(y)} \quad (5)$$

where y_i and $\hat{y}_i$ are the calibration target value and predicted target value of the i -th sample, respectively, and l denotes the total number of samples in the test set. The $\max(y)$ and $\min(y)$ represent the maximum and minimum of the predicted target values in the test set, respectively. After normalization, its values range between 0 and 1. The closer the value is to 0, the more accurate the model prediction; conversely, the closer it is to 1, the poorer the model prediction.

The *MAE* quantifies the average magnitude of the absolute differences between predicted and calibration values, without considering the direction of errors. A smaller *MAE* indicates that the predicted values are closer to the true values. Its definition is as follows³³:

$$MAE = \frac{1}{l} \sum_{i=1}^l |y_i - \hat{y}_i| \quad (6)$$

The *RMSE* reflects the square root of the average of squared differences between predicted and calibration values. It amplifies the impact of large errors, making it more sensitive to outliers. A smaller value denotes higher prediction accuracy. Its definition is³³:

$$RMSE = \sqrt{\frac{1}{l} \sum_{i=1}^l (y_i - \hat{y}_i)^2} \quad (7)$$

The R^2 reflects the degree to which the model interprets the target variable, indicating its goodness of fit to data fluctuations. Its value ranges between 0 and 1. Values closer to 1 indicate superior model fit, characterized by smaller discrepancies between predicted and calibration values, demonstrating enhanced capability to explain data variability. Conversely, values approaching 0 signify poorer model fit and diminished capacity to account for

data variability. The definition of R^2 is given by²⁵:

$$R^2 = 1 - \frac{\sum_{i=1}^l (y_i - \hat{y}_i)^2}{\sum_{i=1}^l (y_i - \bar{y})^2} \quad (8)$$

where $\bar{y}$ denotes the mean of the calibration target value.

The *ET* of machine learning models serves as a critical metric for assessing model efficiency and practical utility²³. Optimizing *ET* not only accelerates model training and prediction but also enhances system responsiveness in practical applications.

Stratified data splitting strategy

In machine learning models, the training set constitutes the data collection used for model training, encompassing both feature labels and corresponding target variables. Through the training set, models establish mapping relationships between inputs and outputs by learning the patterns in the data to adjust internal parameters, thereby minimizing prediction error. While larger training set enhances a model's capacity to capture complex mappings between input feature labels and output target variables, but it is also prone to overfitting. The training set typically comprises 70–80% of the total data. The validation set is an intermediate data partition between the training set and the test set, specifically used for hyperparameter tuning and model complexity optimization, it neither participates in the update of model parameters during the training process nor serves as unseen data for the final generalization evaluation. Its core function is to prevent overfitting caused by improper hyperparameter settings and help select the optimal model structure. The test set serves as the data collection for evaluating model performance and this data has never been used at all during the model training process. Its function is to assess generalization capability of the model, that is, to determine how the model performs on unencountered data.

To address the adjacent sample leakage bias that may occur when randomly splitting data from the same device and batch, and to enhance the rationality of data distribution and generalization reliability, we divided the data into training, validation, and test sets in an 8:1:1 ratio. Considering that temperature has a more significant impact on the sensing performance of SAW sensors, we performed stratified sampling based on temperature intervals. Specifically, we divided the temperature range into 3 equal-frequency intervals, ensuring that the proportion of samples in each temperature interval within the training, validation, and test sets remains consistent. This avoids data distribution imbalance caused by excessive concentration of samples in specific temperature ranges. Finally, we fixed the random seed during the splitting

process to ensure the consistency and reproducibility of data division, eliminating result fluctuations caused by random splitting.

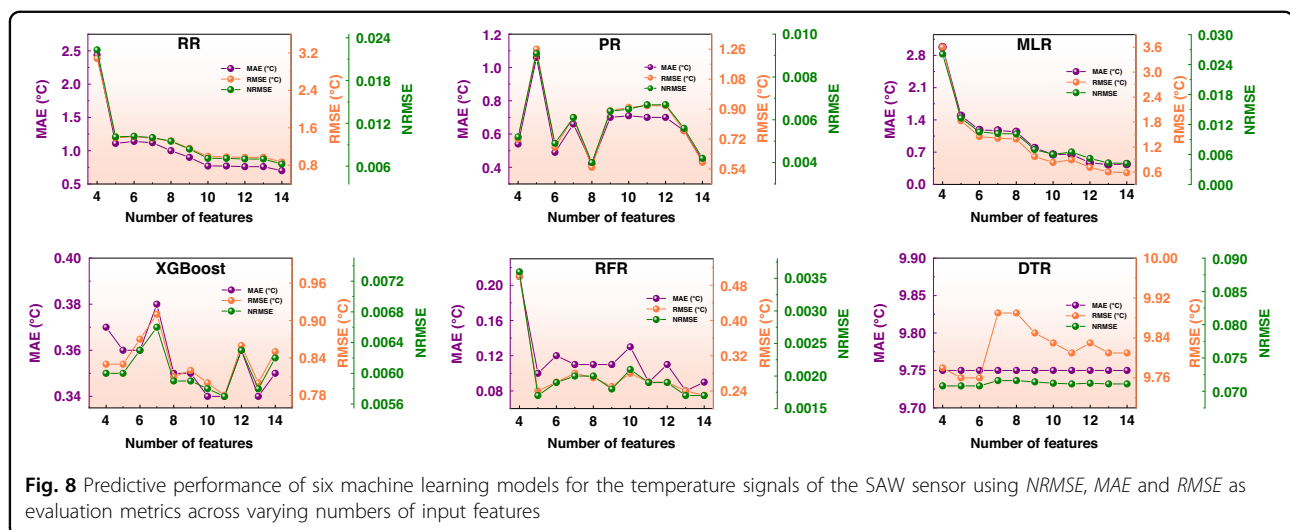
Results and discussion

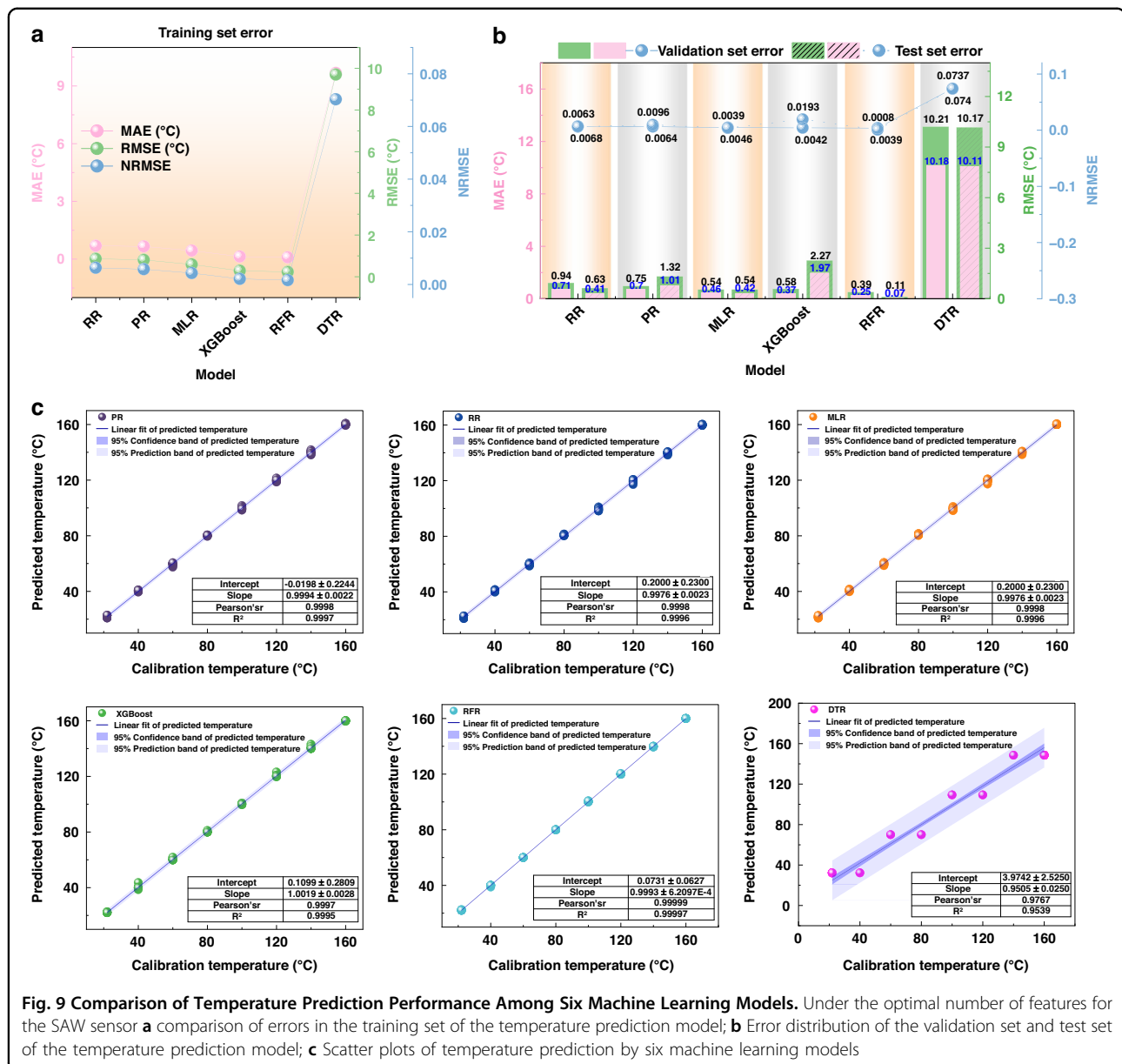
We employed six machine learning models: RR, PR, MLR, XGBoost, RFR, and DTR, for predicting both temperature and strain. Following the feature importance ranking illustrated in Fig. 8, features were sequentially incorporated into model training in descending order of importance. Predictive performance was evaluated using the *NRMSE*, *MAE* and *RMSE* evaluation metrics concurrently. Notably, models demonstrated generally inferior performance with fewer than four input features. Consequently, only results with four or more input features are presented in this study.

Temperature prediction performance

Figure 8 shows the prediction performance of six machine learning models on the temperature signals of SAW sensors. It can be seen from the figure that there are significant differences in the number of features required for various models to achieve optimal prediction results, which intuitively reflects the inherent differences in the logic of feature information capture and processing among different algorithms. When the number of features is less than 4, the RR model has a high prediction error due to insufficient coverage of effective information; as the number of features increases, the dimension of effective information is supplemented, and the prediction error decreases significantly. In addition, thanks to the regularization mechanism of the RR model, which can effectively suppress the interference of redundant features, when the number of features further increases, the prediction error only fluctuates slightly; when the number of features reaches 14, the effective information is

sufficient to support high-precision prediction without overfitting, and the model error reaches the minimum at this time, the *MAE* is 56.39 °C, the *RMSE* is 69.21 °C, and the *NRMSE* is 0.0865. The performance change trend of the MLR model is basically consistent with that of the RR model. When the number of features is 14, its *MAE* is as low as 0.44 °C, and the introduction of redundant features does not significantly increase the model error. The error fluctuation range of the PR model under different numbers of features is relatively large. This is because the change in the number of features directly affects the order and quantity of polynomial terms. That is, if the non-linear correlation between the newly added features and the target variable is weak, the polynomial redundancy will increase sharply, leading to an increase in error; while a reduction in the number of features will cause a contraction in the polynomial dimension, resulting in underfitting of the model and an increase in error as well. The model achieves the optimal prediction effect when the number of features increases to 8, with a corresponding *MAE* of 0.42 °C, *RMSE* of 0.55 °C, and *NRMSE* of 0.004. The XGBoost model can independently mine the correlation between features through the splitting process of decision trees, and its dependence on the number of features is significantly lower than that of linear models. It only needs 10 features to capture the core information of the data. At this time, the *MAE* of the model is 0.34 °C, the *RMSE* is 0.78 °C, and the *NRMSE* is 0.0057. The subsequent addition of redundant features is difficult to further improve the model performance. The performance of the RFR model is particularly prominent. When the number of features is only 5, its *MAE* is as low as 0.1 °C, and the model information quickly reaches a saturated state. The subsequent increase in the number of features only brings a small fluctuation in the error; when the number of features is 13, the *MAE* further drops to





0.08 °C, which is the lowest error among the six models. The prediction error of the DTR model in the entire feature range is at a relatively high level (the MAE is 9.75 °C), and changes in the number of features can hardly improve its prediction performance, which indicates that the fitting ability of this algorithm for such temperature signals is relatively weak.

Figure 9 shows the temperature prediction results of six models under the optimal input feature conditions. It can be seen from Fig. 9a that the RFR model has the best training effect. The MAE and RMSE values of its training set are significantly lower than those of other models, and the NRMSE is close to 0, reflecting its accurate fitting ability to the temperature training data. In addition, it can

be seen from Fig. 9b that the difference between RMSE and MAE of the RFR model is extremely small. Among them, the MAE of the test set is 0.11 °C, the RMSE is 0.07 °C, and the difference between the two is only about 0.04 °C. This means that there are few extreme errors in the prediction results, and the prediction results of the model are more stable. In addition, the error gap between the validation set and the test set of the RFR model is also very small, indicating that the model has strong generalization ability, and the prediction effect on new data is consistent with that in the training stage.

Figure 9c presents the calibration scatter plots of the 6 models in the temperature prediction task, showing the fitting relationship between the predicted temperature and

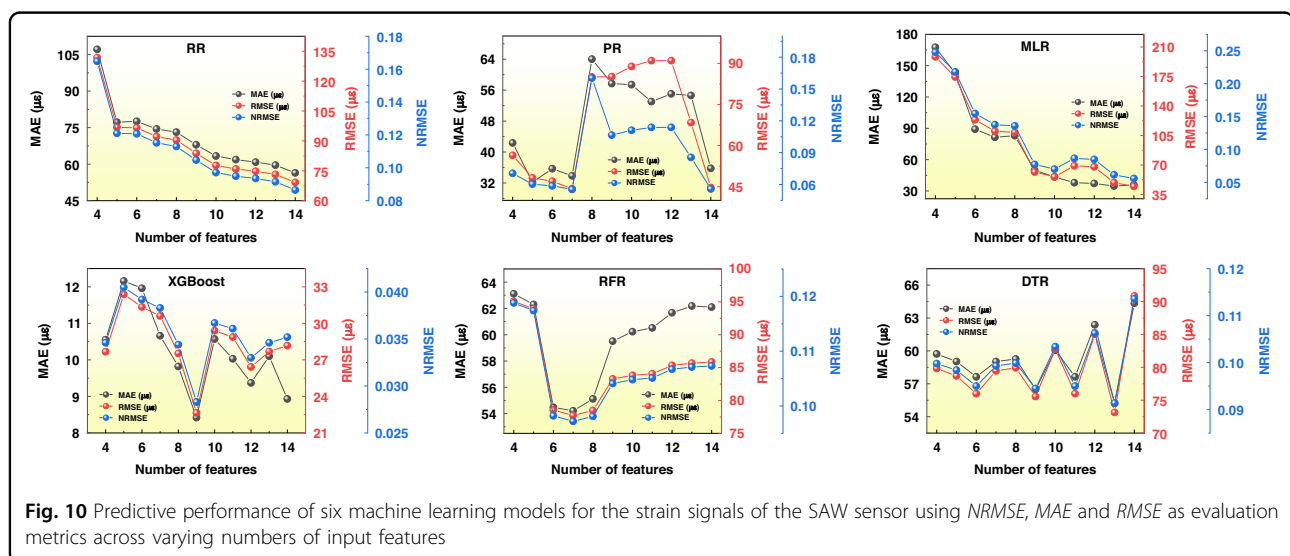
the calibrated temperature of each model. Each data point in the figure corresponds to a set of predicted value—calibrated value pairs. The blue straight line is the linear fitting line of the two, and the blue area represents the 95% confidence/prediction interval. Among them, the 95% confidence interval means that there is a 95% probability that the true fitting line of the model falls within this area, and the 95% prediction interval means that there is a 95% probability that the predicted value of a new sample falls within this area. A narrower interval means lower uncertainty in the model's prediction and more stable results. From the perspective of fitting effect, except for the DTR model, the prediction results of the other models are excellent. Their prediction points are closely fitted to the fitting line, the intercept is close to the ideal value of 0, the slope is close to the ideal value of 1, and both the correlation coefficient and the coefficient of determination reach more than 0.99, indicating a strong consistency between the predicted values and the true values. The fitting performance of the DTR model is slightly weaker. Not only do the prediction points deviate from the fitting line, but the intercept and slope also deviate significantly from the ideal values, and the coefficient of determination is relatively low. Furthermore, in the supplementary materials, we further provide scatter plots with Y error bars corresponding to Fig. 9c, which can intuitively reflect the absolute error between the predicted value of each model and the calibration target value.

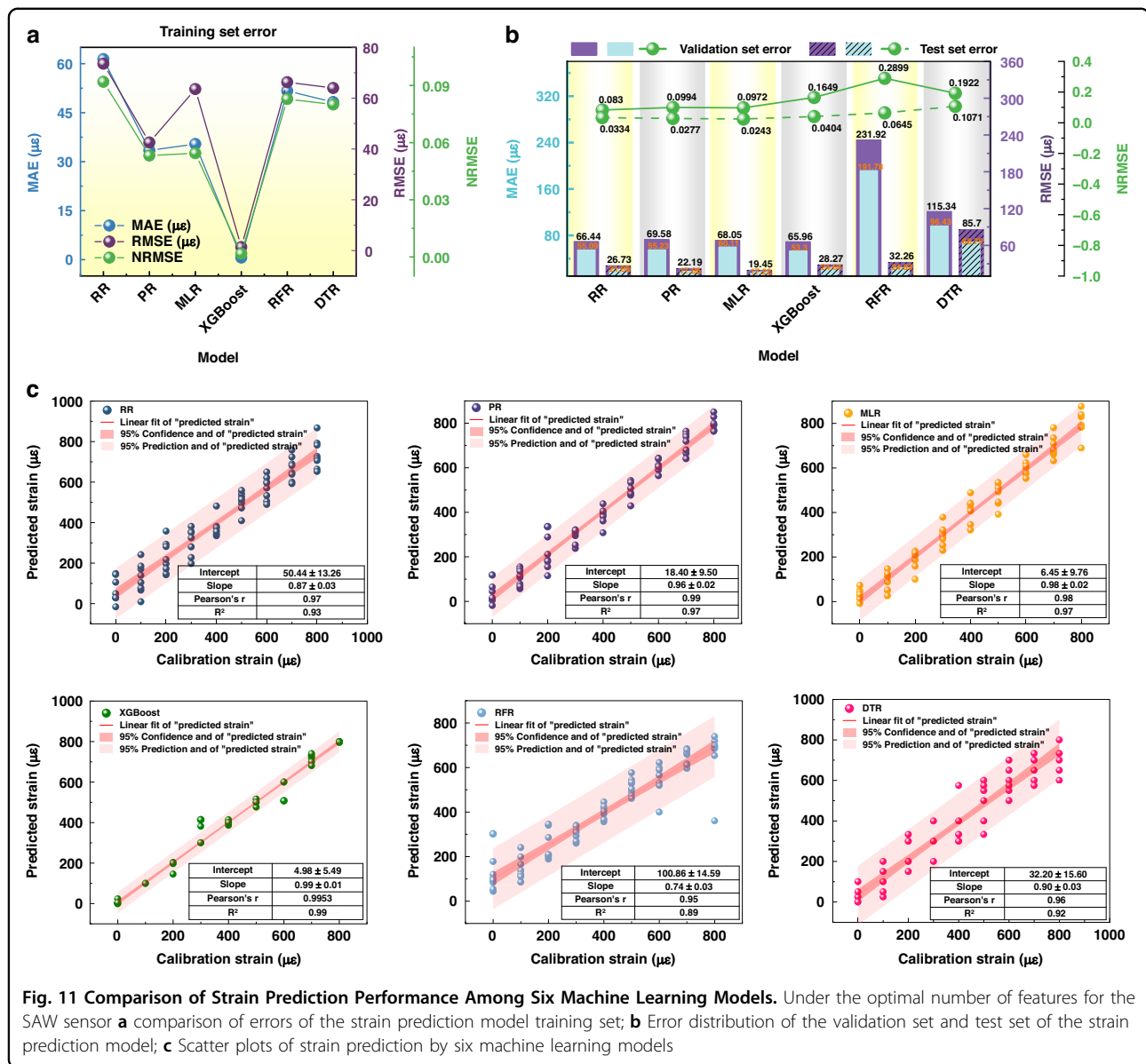
Strain prediction performance

Figure 10 shows the prediction performance of 6 machine learning models on the strain signals of SAW sensors. Since the sensitivity of the sensor to strain is lower than that to temperature, the *NRMSE* of the strain prediction results is generally higher than the deviation

level of temperature prediction. At the same time, the performance of different models shows significant differences. Among them, the RR model and MLR model show cross-task consistency in the strain prediction task, that is, when the number of input features exceeds 4, the model prediction error decreases rapidly, and when the number of features increases to 14, the prediction error reaches the global minimum. However, the prediction error of the PR model still fluctuates under the influence of the number of features, which is mainly because the number of features will change the complexity of the polynomial. The model achieves the optimal performance when the number of features is 7, with a corresponding *MAE* of 33.84 $\mu\epsilon$, *RMSE* of 44.25 $\mu\epsilon$, and *NRMSE* of 0.0553. In contrast, the XGBoost model still shows low dependence on the number of features. When the number of features is 9, its strain prediction error drops to the lowest among the six models, with *MAE* of 8.42 $\mu\epsilon$, *RMSE* of 22.65 $\mu\epsilon$, and *NRMSE* of 0.0283. In addition, the *MAE* of the RFR model has dropped to 54.21 $\mu\epsilon$ when the number of features is 7, but as the number of features continues to increase, redundant information causes overfitting, leading to a gradual rise in the model's prediction error. The DTR model maintains a high prediction error in the entire feature range, indicating that its fitting ability for such strain signals is relatively weak.

Figure 11 shows the strain prediction results of 6 models under the optimal input feature conditions. In terms of fitting effect, the XGBoost model has the best training effect, with the *MAE* of 0.13 $\mu\epsilon$, the *RMSE* of 0.29 $\mu\epsilon$, and the *NRMSE* of 0.0021 for the training set. Since strain prediction involves the coupled nonlinear relationships of multiple factors such as temperature and sensor spectral characteristics, XGBoost, through the integrated structure of gradient boosting trees, can fit





residuals layer by layer and accurately capture the complex mapping between features and strain, which is difficult to achieve with linear models. In addition, as can be seen from Fig. 11b, the XGBoost model also outperforms other models in the validation set and test set, with an *NRMSE* of 0.0459 for the validation set and only 0.0404 for the test set. The numerical difference between the two is extremely small, which not only reflects strong fitting ability but also good generalization stability. Overall, the XGBoost model has more prominent overall prediction accuracy and reliability.

Figure 11c presents the calibration scatter plots of 6 models in the strain prediction task, showing the fitting relationship between the predicted strain and the calibrated strain of each model. The red straight line is the

linear fitting line of the two, and the pink area represents the 95% confidence/prediction interval. In terms of fitting effect, the XGBoost model performs the best. Its prediction points are closely the fitting line, with the intercept of 4.98 is close to the ideal value of 0, and the slope of 0.99 is close to the ideal value of 1, and both the correlation coefficient and the coefficient of determination reaching above 0.99, indicating a very strong consistency between the predicted values and the true values. The PR and MLR models are the next, with both having a coefficient of determination of 0.97, and their fitting effects are relatively stable. The fitting performance of the RR and DTR models is slightly weaker, with coefficients of determination of 0.93 and 0.92 respectively, and the dispersion degree of the prediction points is slightly higher. The RFR

Table 3 *ETs of 6 machine learning models for predictions of temperature and strain signals in the SAW sensor under optimal input feature conditions*

Models	ET of temperature prediction (ms)	ET of strain prediction (ms)	Input features
RR	1	1	$f_n, B_w, S_{11}, Q, f_{min}, f_{max}, S_{11} \text{ skewness, mean } S_{11}, f_r^2, B_w^2, S_{11}^2, Q^2, f_r * S_{11}, B_w * S_{11}$
PR	1	1	$S_{11}, f_r * S_{11}, f_r^2, B_w, f_n, B_w^2, Q$
MLR	1	1	$B_w, f_n, f_{min}, f_{max}, Q^2, B_w * S_{11}, Q, S_{11}^2, S_{11} \text{ skewness, mean } S_{11}, f_r * S_{11}, B_w^2, S_{11}, f_r^2$
XGBoost	1566.6	595.8	$Q^2, S_{11} \text{ skewness, } f_n, f_{min}, \text{mean } S_{11}, Q, B_w * S_{11}, S_{11}, S_{11}^2$
RFR	209.9	243.5	$S_{11} \text{ skewness, } f_r^2, f_{min}, f_n, f_{max}, B_w * S_{11}, B_w$
DTR	1	1	$S_{11} \text{ skewness, } f_n, f_{max}, B_w, S_{11}, B_w * S_{11}, \text{mean } S_{11}, B_w^2, f_r^2, Q^2, f_r * S_{11}, f_{min}, S_{11}^2$

model performs the worst: not only do the prediction points obviously deviate from the fitting line, but the intercept of 100.86 and the slope of 0.74 both deviate significantly from the ideal values, and the coefficient of determination is only 0.89. The core difference in this result stems from the characteristics of the models. XGBoost achieves accurate modeling through the integration strategy of layer-by-layer fitting of residuals and can effectively capture the complex relationships between features. In contrast, linear models such as RR and MLR can only fit linear relationships and are difficult to handle nonlinear interactions of features. The RFR model has insufficient generalization ability and the DTR model has limited fitting accuracy, which ultimately leads to poor prediction effects. For more details on the scatter plot of strain predictions with Y error bars, please refer to the supplementary materials.

Table 3 compares the ETs of six machine learning models for predictions of temperature and strain signals in SAW sensors under optimal input feature conditions. It can be seen that compared with models such as RR, PR, MLR, and DTR, which take only about 1 ms for prediction, the prediction time of XGBoost is relatively longer (1566.6 ms for temperature prediction and 595.8 ms for strain prediction). However, in this study's task of predicting temperature and strain signals of SAW sensors, the XGBoost model has shown outstanding advantages in prediction accuracy and is the model with the best comprehensive performance in terms of accuracy in this experiment. Overall, although this model is slightly insufficient in execution efficiency, its excellent prediction accuracy makes it more suitable for application scenarios with high accuracy requirements and relatively loose real-time requirements.

Conclusions

In this work, we developed an LN-based single resonator-type SAW sensor and broadened its operational temperature to 600 °C by introducing an AlN protective

layer. Under environments with no temperature fluctuations or strain interference, the sensor exhibits excellent strain/temperature sensing characteristics. To address the cross-sensitivity issue between temperature and strain signals in SAW sensors, we introduce machine learning technology. By extracting the features of S_{11} spectra under varying temperature and strain loading conditions, machine learning algorithms are used to establish mapping relationships between input feature labels and output target variables, thereby enabling temperature/strain prediction. We investigate the performance of six machine learning models in predicting temperature/strain under different input feature conditions. Experimental results demonstrate that the XGBoost model achieved the best overall prediction performance. The R^2 values for temperature and strain reached 99.97% and 99.23% respectively, the $NRMSEs$ were as low as 0.0283 and 0.0059 respectively, the $RMSEs$ were 0.82 °C and 22.65 $\mu\epsilon$ respectively, and the $MAEs$ were 0.35 °C and 8.42 $\mu\epsilon$ respectively. The research realizes high-precision temperature-strain composite sensing under complex environmental conditions, and provides a new method for multi-parameter information acquisition using monolithic SAW sensors. In future work, this method can be extended to directions such as multi-parameter decoupling, models adapted to lightweight edge computing, dynamic condition robustness optimization, and spatial multi-variable inversion based on sensor arrays, so as to further improve the universality and applicability of the SAW sensing system.

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Conflict of interest

The authors declare no competing interests.

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